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SENG1050

Web Technologies

Week 1 - The Internet: Concepts and Communication

Lecture Outline

- ❑ Internet and Network
- ❑ Internet Protocols
- ❑ World Wide Web (WWW)

Internet & Network



The Internet

- ❑ A global network connecting billions of computing devices.
- ❑ A **network of networks**.
- ❑ Constantly growing and changing.

Network

A group of computing devices that are connected together to permit the (rapid) transfer of information between them.



Networks: Categories

Local Area Network (LAN)

- + Cheap and easy to install.
- + Geographically close (easy to maintain & secure).
- + Controlled at the department/group level.

Wide Area Network (WAN)

- + Effective for large transnational companies.
- + WAN transmissions don't burden the local LANs. They route through special purpose computers.
- Expensive : costs of planning, infrastructure, hardware, etc.
- Different LAN/WAN technologies are incompatible.



Network: LAN Physical Topologies

Bus: all computers share a single line of communication; need a way to avoid/detect collisions.

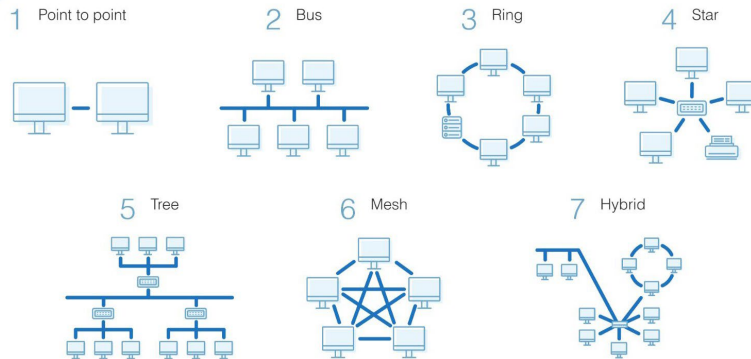
Ring: each computer communicates only with its neighbours, forming a cycle or *ring*; information is relayed in one direction.

Star: all computers communicate via a central *switch/hub/router*.

Tree: a bus topology with branches.

Mesh: every computer has a direct line to every other computer.

Network Topology Types



Which LAN topology is most commonly used in homes and small business?

Bus

0%

Ring

0%

Star

0%

Tree

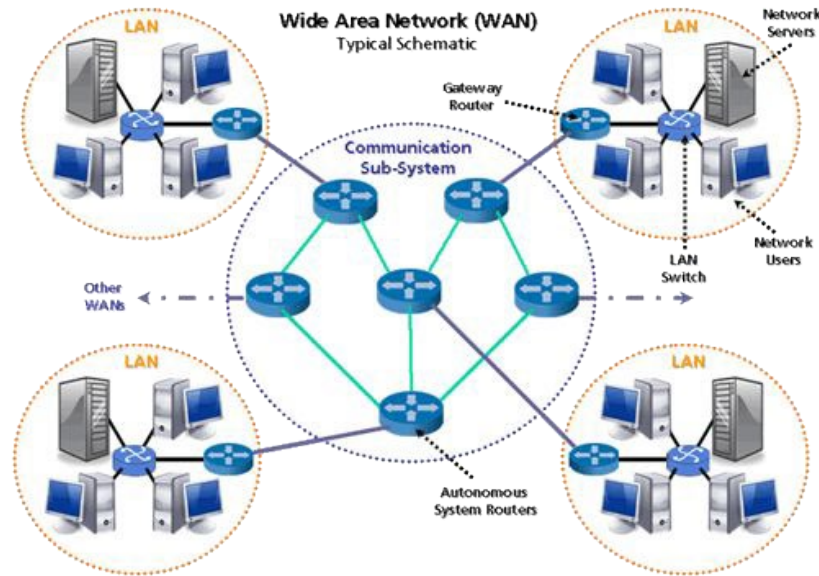
0%

Mesh

0%



Networks: Wide Area Networks (WAN)



Source: <http://www.air-stream.org.au/network/what-wide-area-network>

Special Networks: Intranet / Extranet / VPN

Intranet

- ❑ Like the Internet, but only accessible from within a certain area (such as a home/office network).
- ❑ A private network, usually owned by a company, and used to provide information and applications for its employees.

Extranet

- ❑ The parts of a company's network that can be accessed from outside.
- ❑ Often restricted to suppliers, vendors, and/or customers
- ❑ Can include webpages and websites.

Virtual Private Network (VPN)

- ❑ A way for a computer to securely access a company's intranet from outside via the Internet.

Networks: Where does the Internet fit in?

The Internet is a network of otherwise incompatible networks.

- ❑ Joins together many LANs and WANs into a single network.
- ❑ Allows computing devices to be connected together and communicate with each other.
- ❑ Based on a set of **agreed upon standards** that a computing device must understand.

Internet Backbone

- ❑ One of the core components of the Internet, where the largest and fastest networks are linked together with fibre-optic connections and high-performance routers.
- ❑ Backbone links carry the majority of the traffic - 'information superhighway'

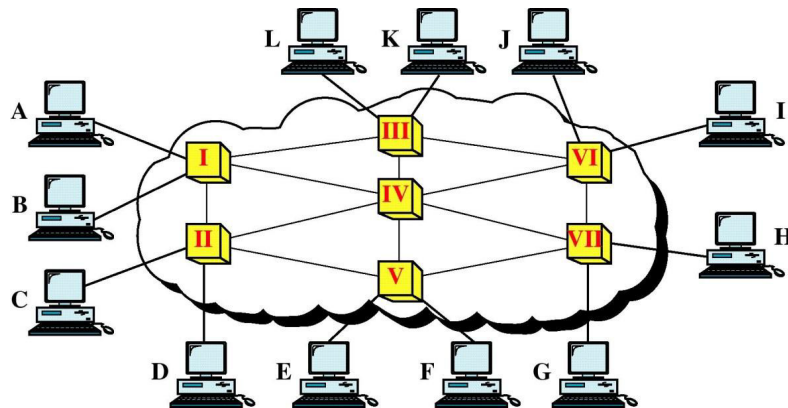


Switching

Point to point connection between each pair of devices is impractical.

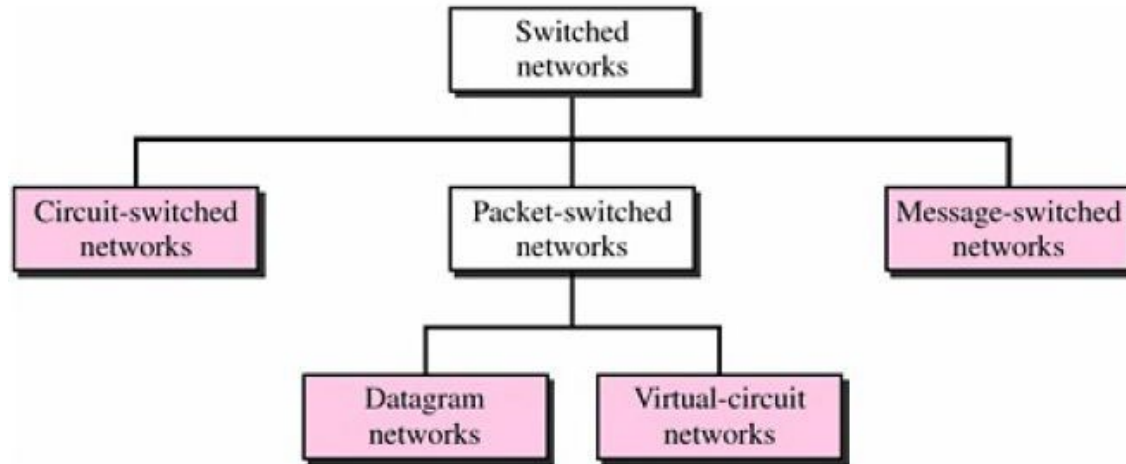
A better solution is **switching**

- ❑ Communicating via a series of interlinked nodes called switches/routers.





Switching: Taxonomy of Switched Networks





Switching

Circuit Switching

A direct connection is established from the sender to the receiver.

Packet Switching (Datagram)

Message is split into packets which are relayed; no guarantee of delivery

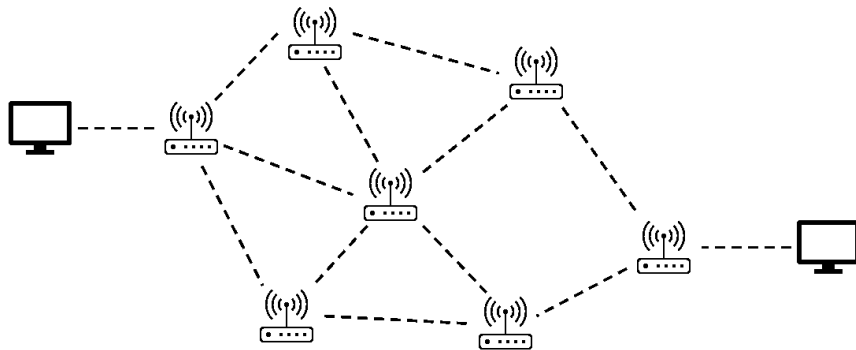
Packet Switching (Virtual Circuit)

Simulating circuit switching using packets

Message Switching

The entire message is relayed; no guarantee of delivery

- ❖ The Internet uses the datagram approach to packet switching.



Circuit Switching

A direct connection is established from the sender to the receiver.

Resources at each node are dedicated to maintaining the connection.

Good for long-distance communication that needs to be maintained for a long period, e.g. landline phones.

Advantages:

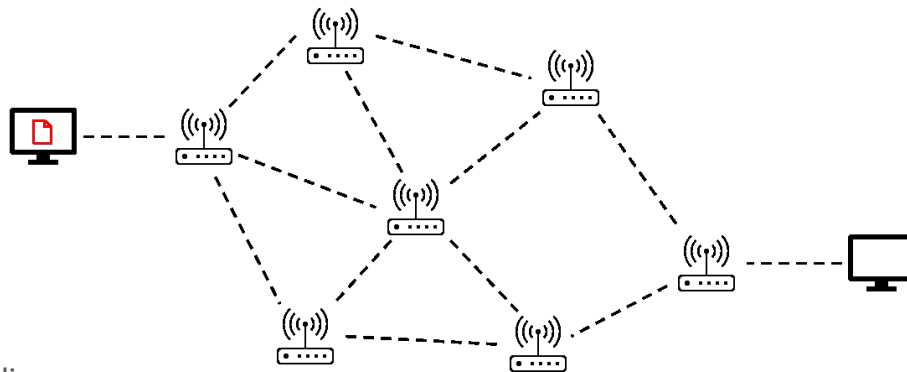
- ❖ Dedicated path provides high reliability and speed
- ❖ Data is guaranteed to arrive in the same order it was sent
- ❖ No delay at intermediate nodes

Disadvantages:

- ❖ Resources reserved cannot be used for anything else during transmission, which is inefficient in high traffic networks, it is expensive to reserve dedicated network resources.
- ❖ Time spent on the initial connection setup can be long
- ❖ Connection path is blocked to other traffic during transmission.



Message Switching



The entire message is sent, no guarantee of delivery.

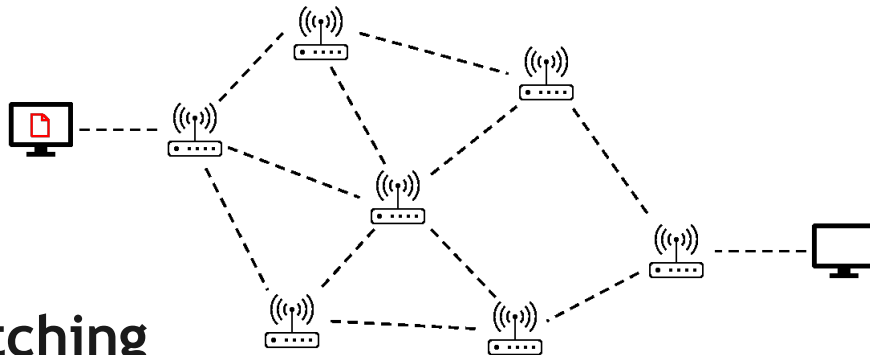
Each intermediate node stores the entire message until the next node is ready (this is called store and forward)

Advantages:

- ❖ Transmission path is not blocked, allowing other devices to share the channel
- ❖ Reduces network congestion, as nodes can store messages until the path to the next node is less busy
- ❖ If an intermediate node fails, the message can be re-routed via a different path.

Disadvantages:

- ❖ Much higher storage requirements for intermediate nodes, as they must store the entire message during each transmission.
- ❖ Very slow, because each node must receive the entire message before forwarding it to the next node.



Virtual-Circuit Packet Switching

Simulates circuit switching using packets.

The message is broken into packets, the first packet determines the route.

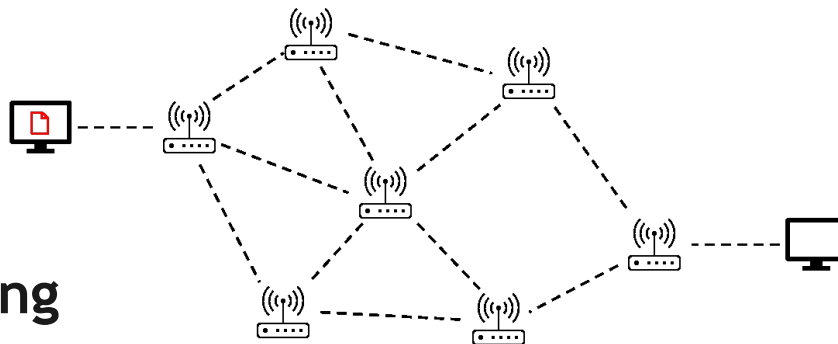
Resources are reserved at each node to receive packets, but does not block the path itself.

Advantages:

- ❖ Transmission path is not blocked, allowing other devices to share the channel
- ❖ Delivery is guaranteed, packets will be received in the same order they were sent.
- ❖ Does not require a large amount of storage space.

Disadvantages:

- ❖ Processing resources (CPU) are reserved at each intermediate node and cannot be used by other network traffic (inefficient on high-traffic networks).
- ❖ Setup time required to establish the route can be long
- ❖ Requires more complex protocols for transmission.



Datagram Packet Switching

Message is split into packets which are relayed – no guarantee of delivery.
Each packets is treated independently, may take different paths from source to destination.

Advantages:

- ❖ No resources are blocked, good for high-traffic networks, multiple devices can share each path.
- ❖ Cheaper to implement
- ❖ If an intermediate node fails, packets can be re-routed via another path.

Disadvantages:

- ❖ Packets can be lost or dropped during transmission
- ❖ Packets are likely to arrive out of order, added header on each packet needed to reassemble correctly
- ❖ Message cannot be interpreted until all packets have arrived, inefficient for communications that require real-time reception such as phone calls.

Forwarding & Routing

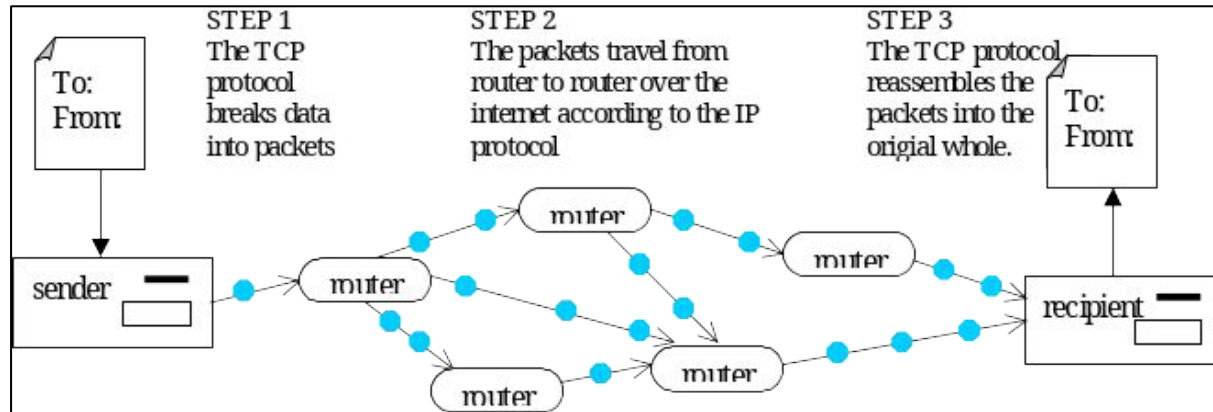
The Internet uses routers to connect networks.

Router

- A device connected to several networks (*building blocks of the Internet*).
- Allows different types of LANs to be connected to the Internet.
- Has two key functions: forwarding and routing.

Routing

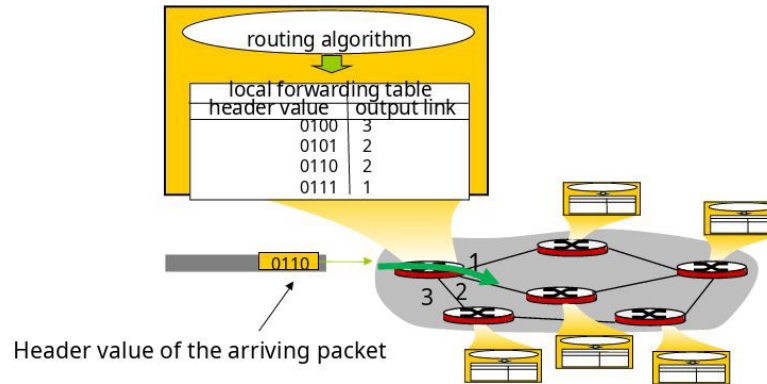
A router runs a routing algorithm to determine the route taken by packets from the source to the destination.



Forwarding

Moving packets from a router's input to the router's appropriate output.

- ❑ Router uses a header and a forwarding table to determine where (i.e., which router output) to forward the packet to.

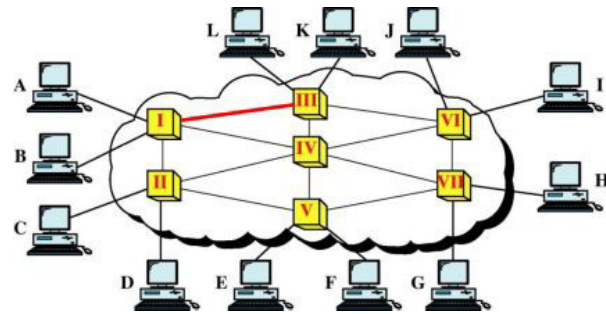


Kurose and Ross (2012)

Overrun

Example:

- ❑ The capacity of the connection between route I and III is 5000 packets.
- ❑ Router I is receiving 3000 packets from host A that it must send to router III.
- ❑ Router I is also receiving 3000 packets from host B that it must send to router III.
- ❑ Unfortunately, the I-III connection can only handle 5000 packets.



- ❑ What happens - a traffic jam?
- ❑ But electrical signals are not like cars!
- ❑ The extra packets will be discarded by the router.
- ❑ The loss must be detected and the packets resent.

Which type of switching should be used for a high-traffic network, where cost needs to be minimized, and the reliability of individual nodes cannot be guaranteed?

Circuit Switching

0%

Message Switching

0%

Virtual Circuit Packet Switching

0%

Datagram Packet Switching

0%



Internet Protocols

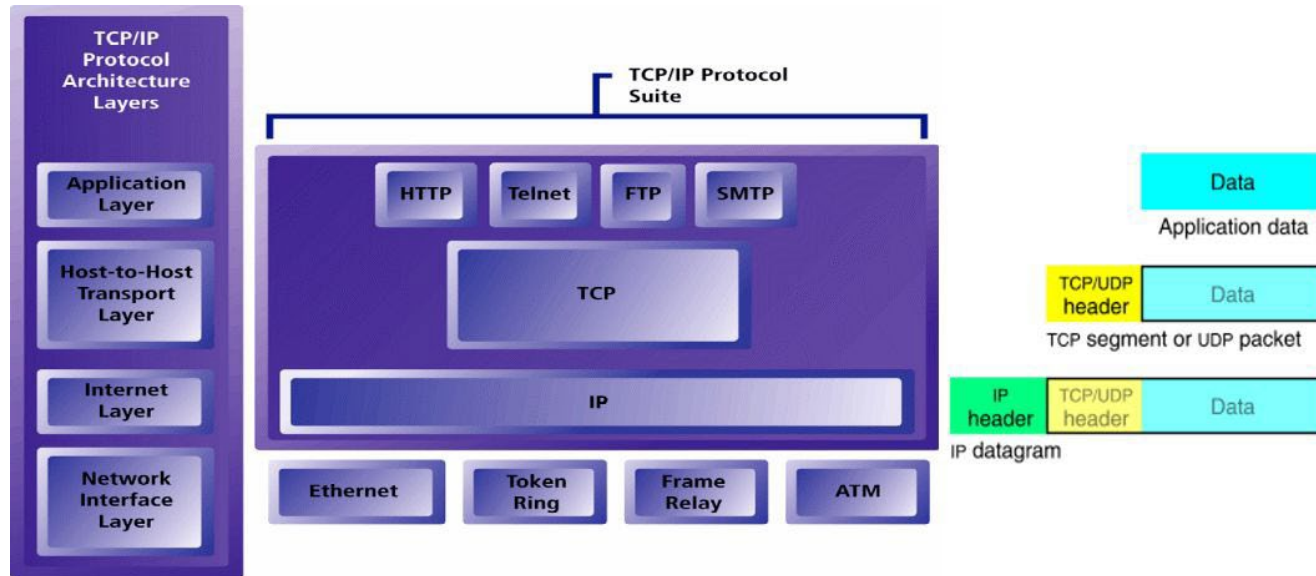
Networks: How does the Internet work?

- ❑ The Internet connects many otherwise incompatible networks into a single virtual network.
- ❑ It does this by defining a set of protocols that networks must use to become part of the Internet.

Protocol

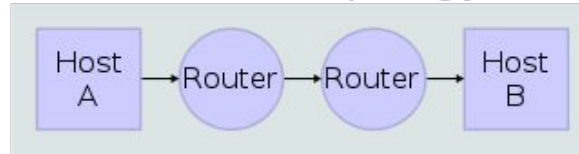
- ❑ Defines a set of rules for communication.
- ❑ Maybe implemented by hardware, software, or combination of both
- ❑ The set of protocols driving the Internet is sometimes called the *Internet Protocol Suite*

The TCP/IP Architecture and Protocol Suite

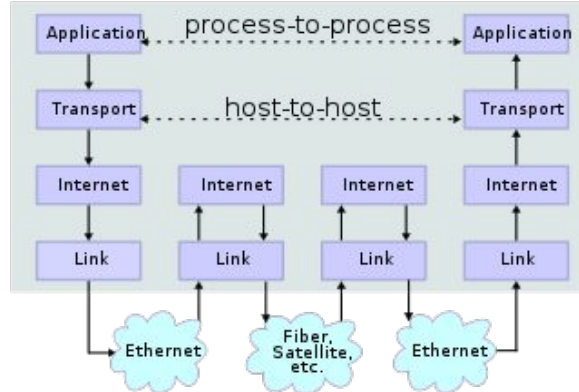


TCP/IP Concepts

Network Topology



Data Flow

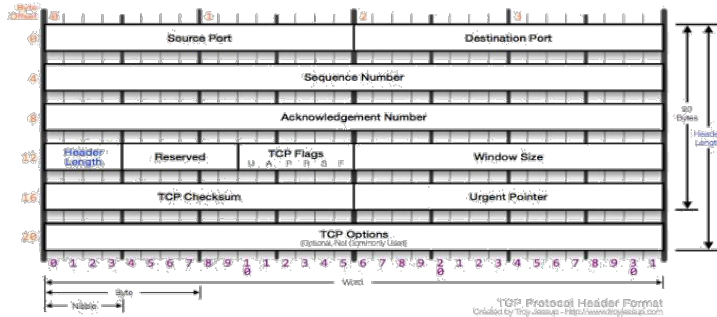




Transmission Control Protocol (TCP)

- ❑ Provides reliable communication over packet-switched networks.
- ❑ Enables two computers (hosts) to establish a connection and exchange streams of data.
- ❑ Breaks up the data into packets.
- ❑ Guarantees the delivered packets are received in the same order in which they were sent.
- ❑ Provides control for rate of transmission.
- ❑ Tells how big a datagram it can handle during the connection creation.

TCP Header



Field Purpose

Source Port Identifies originating application
Destination Port Identifies destination application

Sequence Number used by receiver to sort datagrams into the right order

Acknowledgment # if the sender doesn't get this within a reasonable amount of time, it sends the data again

Header Len Length of TCP header

Flags TCP flags: SYN, FIN, RST, PSH, ACK, URG

Window Control how much data can be in transit at one time

Checksum "sum" of headers bytes - used to detect errors.

Urgent Pointer eg., Ctrl-C

Options Special TCP options such as MSS and Window Scale

Ports and Protocols

- ☐ 20 FTP
- ☐ 22 SSL
- ☐ 23 Telnet
- ☐ 25 SMTP
- ☐ 53 DNS
- ☐ 80 HTTP
- ☐ 8008, 8080 alternate ports for HTTP
- ☐ 443 HTTPS

- Each application communicating on the network will use its own port.
- Ports 0 – 1023 are reserved for system processes and common protocols by IANA (Internet Assigned Numbers Authority)
- User application software can use ports 1025 – 65535.



Internet Protocol (IP)

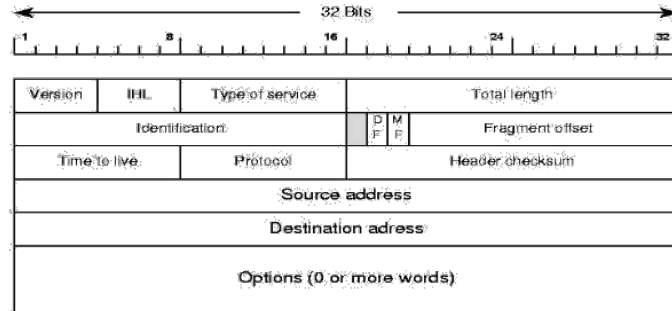
Rules define how packets are routed across the Internet:

- ☐ How to specify source and destination.
- ☐ How to get from A to B.

A packet switching protocol used on the Internet.

- ☐ Sends data in individual packets.
 - ☐ A packet conforming to the IP protocol is called an IP Datagram.
- ☐ Maximum size of a packet is determined by the networks.
 - ☐ Fragmented if too large.
- ☐ Unreliable.
 - ☐ Packets might be lost, corrupted, or delivered out of order.

IPv4 Datagram Header



Field Purpose

Vers IP version number

Len Length of IP header

TOS Type of Service

Total Len Length of entire datagram

Ident. IP datagram ID (for frag/reassembly)

Flags Don't/More fragments

Frag Off Fragment Offset

Field Purpose

TTL Time To Live - Max # of hops

Protocol Higher level protocol (1=ICMP, 6=TCP, 17=UDP)

Checksum Checksum for error detecting

Source IA Originator's Internet Address

Dest. IA Final Destination Internet Address

Options Source route, time stamp, etc.

Data... Higher level protocol data

This will be covered further in week 10



IP Addressing

IP address:

- ❑ A 32-bit identifier that uniquely and universally defines a device's connection to the Internet.
 - ❑ Approx 4.29 billion unique addresses.
- ❑ Each host on the Internet has an IP address.
- ❑ Usually expressed as four period-separated decimal numbers (each is 8 bits, i.e., 0 - 255).
 - ❑ Example: 13.248.219.129 (www.duckduckgo.com)



N.B: Binary is read from right-to-left, each bit increases the power of 2. e.g.

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	0	0	1	1	0	0	1
10011001 = 153							

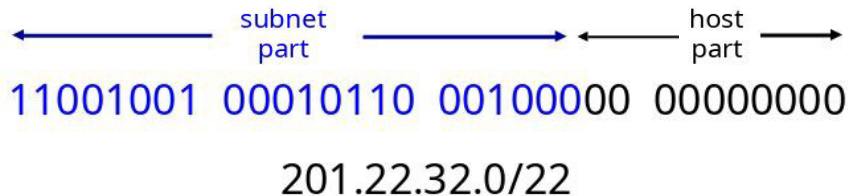
IP Addressing

An IP address has two parts:

- ❑ #1: an Internet *network number*.
- ❑ #2: the *local number* of a particular device (server, workstation, printer, etc) within that network.

The common classless addressing is CIDR (**C**lassless **I**nter**D**omain **R**outing)

- ❑ Address format: **a.b.c.d/x**, where:
 - ❑ **a.b.c.d** defines an address
 - ❑ **x** defines # bits in the subnet portion of the address



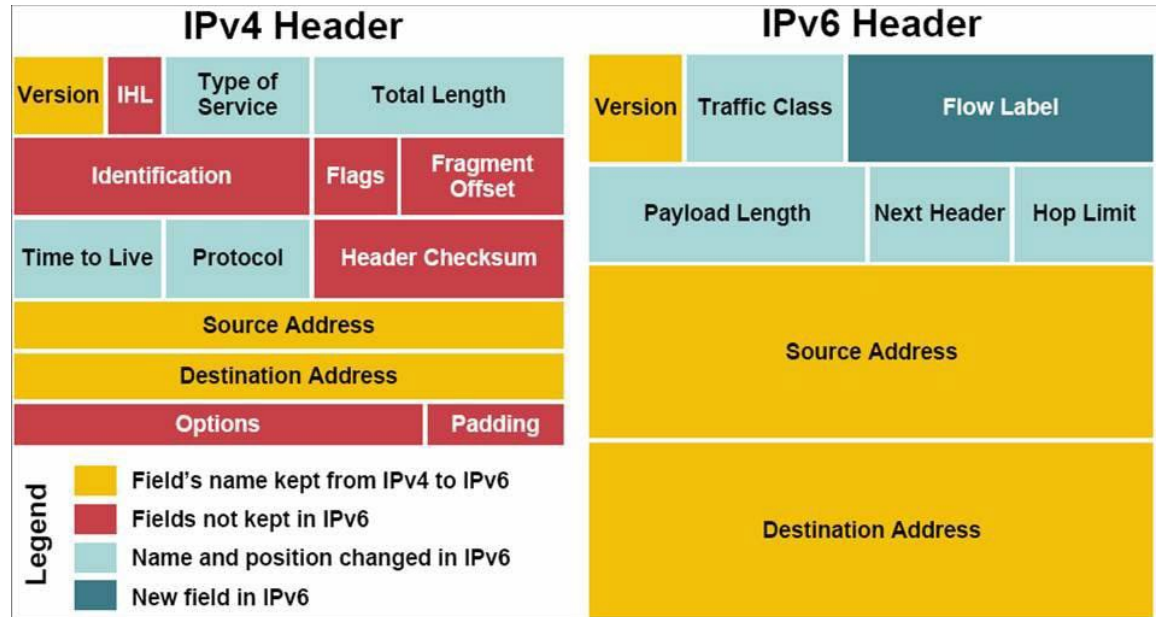
What happens when we run out of IP addresses?

- ❑ This problem was identified many years ago.
- ❑ Adjustments were made to the IPv4 protocol to allow for an **endless** number of computing devices.
- ❑ The new standard is called IPv6, which uses 128-bits to identify each device instead of 32-bits.
- ❑ IPv4 and IPv6 can work together
- ❑ Users are gradually converting from IPv4 to IPv6, and IPv6 will be the default soon.

IPv6 provides a maximum of 2^{128} hosts, i.e., approximately 3.4×10^{38}

- ❑ Which means, *"there are 5 million addresses per square inch on the sphere that encloses the solar system out to Pluto."*
 - ❑ (<https://www.teracomtraining.com/tutorials/teracom-tutorial-how-many-ipv6-addresses-are-there.htm>)

IPv6 Header



A switch is a network device that performs packet forwarding based on destination MAC address. Which layer of the TCP/IP protocol stack is it operating in?

Application layer

0%

Transport layer

0%

Internet/Network layer

0%

Network access/Data link layer

0%





Domain Name System (DNS)

DNS provides a symbolic equivalent for numeric IP addresses

- ❑ 20.43.111.112 => duckduckgo.com
- ❑ Each computer knows (at least) one local DNS server, which human memorable domain names to numeric IP addresses.

Each local server knows the IP number <-> name mappings for its local network.

If the requested mapping is outside of the **local domain**, then the **local server** asks a **root server**, which relays the information to the appropriate local server.

Simple Mail Transfer Protocol (SMTP)

Email is one of the most popular Internet applications.

- ❑ The first email was sent in 1971 by Ray Tomlinson.

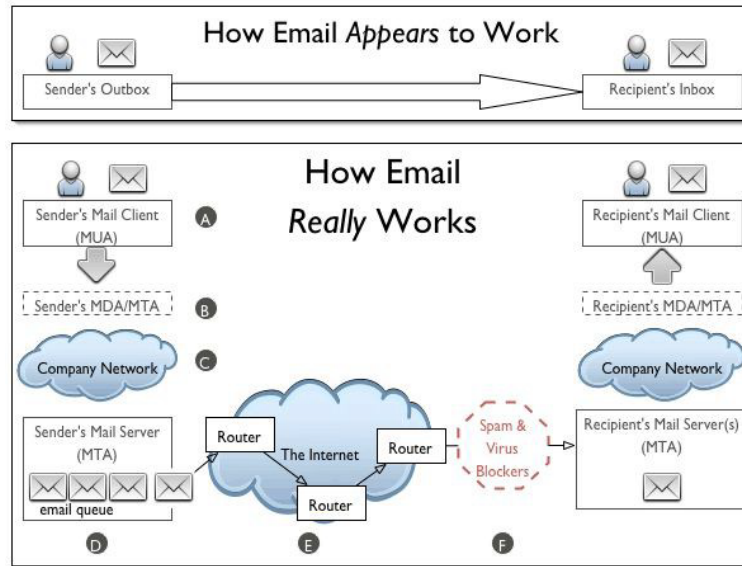
The Simple Mail Transfer Protocol (SMTP) is a protocol for sending emails over the Internet.

- ❑ SMTP conforms to the TCP/IP (packets, etc)
- ❑ Delivery time under SMTP is not guaranteed.

Email systems are based on store-and-forward model.

- ❑ Email servers accept, forward, deliver, and store messages.
- ❑ Sender and recipient don't have to be simultaneously online.

What happens to Email?





HyperText Transfer Protocol (HTTP)

- ❑ A set of standards allowing internet users to exchange website information.
- ❑ There have been several HTTP iterations since its introduction in 1991.



- ❑ Generally run over TCP/IP
 - ❑ The World Wide Web (WWW) came long after the Internet
- ❑ Requests are addressed using **Uniform Resource Identifiers (URIs)**

N.B. There is also HTTP 3.0, published as a fully-fledged web standard in 2022 – currently supported by less than a third of active websites.



HTTP/1.x

- ❑ Used in hypermedia systems distributed across the network.
- ❑ Message types: requests and responses
 - ❑ **Requests** are made to get or send information.
 - ❑ **Responses** are replies to request messages.



HTTP/1.x: Request Line



Example: `GET /path/to/file/index.html HTTP/1.0`

Kinds of Requests

- ❑ **GET:** returns whatever information is requested
- ❑ **POST:** includes a body message. For example: sending *forms* to the server
- ❑ **HEAD:** returns the *Request-URI* header information.
- ❑ Also PUT, DELETE, TRACE, OPTIONS, and PATCH.



HTTP/1.x: Response

First line of Full-Response is the Status-Line

Example: HTTP/1.1 200 OK



1xx: Informational

2xx: Success

- The action was successfully received, understood, and accepted.

3xx: Redirection

- Further action must be taken to complete the request

4xx: Client Error

- The request contains bad syntax or cannot be fulfilled
- 401 Unauthorized - you need a password!
- 403 Forbidden
- 404 Not found - file doesn't exist (or 403!)

5xx: Server Error

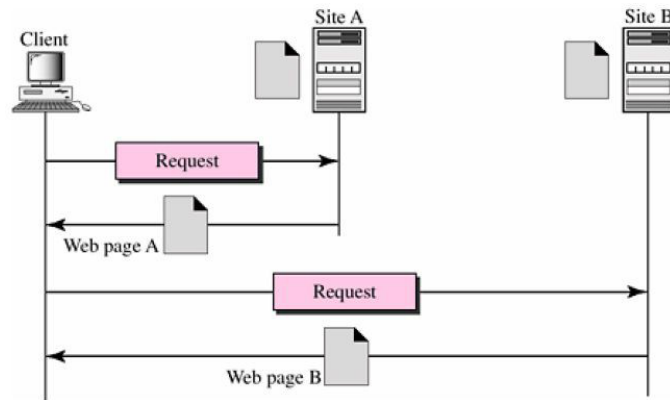
- The server failed to fulfill an apparently valid request
- 500 Internal Server Error

World Wide Web (WWW)



The World Wide Web (WWW)

- ❑ The WWW is a distributed client/server service, in which a client using a browser can access a service from a server.
- ❑ The service provided is distributed over many locations - called sites.
- ❑ Each site holds one or more Web pages.
- ❑ Based on hypertext - instant cross-referencing: links from a word, phrase or image to other pages.
- ❑ A web browser is commonly used to connect to a website's server and display its web pages; but there are also other ways to access this information.





The World Wide Web (WWW)

Initial idea by Tim Berners-Lee.

📄 <http://www.w3.org/People/Berners-Lee/Overview.html>

Has three aspects:

1. A scheme for locating distributed documents
 - **URI**: URL or URN
2. A protocol for accessing these documents
 - **HTTP**
3. A HyperText language linking the documents together
 - **HTML**



Summary

- ❑ We defined our goal by learning about the internet, websites, and how a website can be 'on' the internet.
- ❑ We learned that the internet is a network of networks
- ❑ We learned that websites are accessed using the world wide web, which led us to refine our goal – rather than just creating a website on the internet, it also needs to be accessible using the world wide web.



Summary

- ☐ Different types of computer networks (LAN, WAN)
- ☐ Network topologies (Bus, Ring, Mesh, Star, Tree)
- ☐ Internet & WWW
- ☐ Packet Switching
- ☐ Forwarding & Routing
- ☐ Layers of TCP/IP Architecture
- ☐ How TCP and IP work together to transfer data
- ☐ IP addressing
- ☐ The role of Domain Name System (DNS)
- ☐ How Email works
- ☐ The working principle of HTTP